

LONDON SOUTHEND, United Kingdom

WMO: 036913

Lat: 51.571N

Lon: 0.696E

Elev: 15

StdP: 101.15

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) -1.2	(d) -5.9	(e) 2.3	(f) 0.2	(g) -4.8	(h) 2.5	(i) 0.3	(j) 13.4	(k) 9.9	(l) 12.2	(m) 9.9	(n) 2.0	(o) 280	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	26.9	19.1	25.0	18.3	23.2	17.1	19.9	25.2	18.8	23.5	18.0	22.3	4.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	12.9	22.2	17.0	12.2	21.4	16.1	11.5	20.4	56.7	25.0	53.5	23.5	50.9	22.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.7	8.8	-5.4	30.5	1.8	1.6	-6.7	31.6	-7.7	32.5	-8.8	33.4	-10.1	34.5
			-6.0	22.2	1.6	1.4	-7.1	23.3	-8.1	24.1	-9.0	24.9	-10.2	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	5.2	5.4	7.1	9.4	12.6	15.7	18.2	18.1	15.3	12.1	8.3	5.8		
	DBStd	5.53	3.33	3.29	3.10	3.02	2.88	2.67	2.44	2.32	2.43	2.77	3.17	3.58		
	HDD10.0	653	151	131	98	46	10	0	0	0	0	12	69	136		
	HDD18.3	2696	408	362	348	268	179	86	33	32	96	194	302	388		
	CDD10.0	1066	2	3	8	28	89	172	254	251	158	77	17	6		
	CDD18.3	66	0	0	0	0	1	7	29	25	4	1	0	0		
	CDH23.3	372	0	0	0	1	5	51	159	138	18	1	0	0		
	CDH26.7	63	0	0	0	0	1	7	26	28	2	0	0	0		
(14) Wind	WSAvg	4.1	4.6	4.7	4.3	3.8	3.9	3.8	3.8	3.8	3.7	4.0	4.1	4.4		
(15) Precipitation	PrecAvg	504	39	29	36	36	35	48	40	47	47	50	52	45		
	PrecMax	659	121	62	85	66	67	110	103	108	165	130	101	95		
	PrecMin	331	11	6	3	4	1	5	8	10	3	2	11	13		
	PrecStd	74	21	15	22	17	17	28	23	23	37	33	21	24		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.2	16.2	21.0	23.9	27.2	29.2	29.9	25.2	20.9	16.2	14.2		
		MCWB	10.9	11.0	11.2	14.4	17.1	19.5	19.6	20.1	18.2	16.4	13.9	11.9		
	2%	DB	12.2	12.2	14.8	18.0	21.2	24.8	26.9	26.8	23.0	18.8	15.1	13.1		
		MCWB	10.1	9.6	10.6	12.5	15.5	18.0	18.9	19.2	17.0	15.3	12.9	11.1		
	5%	DB	11.1	11.2	13.1	16.1	19.8	22.8	25.0	24.8	21.0	17.2	14.0	12.2		
		MCWB	9.2	8.7	9.7	11.4	14.6	16.8	18.1	18.4	16.2	14.4	11.8	10.2		
	10%	DB	10.1	10.1	12.0	14.8	18.0	21.0	23.2	22.9	19.9	16.2	13.0	11.1		
		MCWB	8.3	8.0	9.1	10.5	13.3	15.8	17.1	17.0	15.5	13.7	10.8	9.2		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.5	12.1	15.0	17.9	20.6	21.0	21.2	19.4	16.9	14.6	12.5		
		MCDB	13.1	13.7	15.4	19.9	22.2	26.6	27.0	26.9	24.0	19.4	16.1	14.2		
	2%	WB	10.7	10.3	11.2	13.1	16.1	18.5	19.7	19.9	17.9	15.7	13.1	11.6		
		MCDB	12.2	12.3	13.7	17.0	20.4	23.5	25.2	25.3	21.5	17.8	14.8	13.2		
	5%	WB	9.7	9.3	10.4	11.9	15.0	17.4	18.7	18.8	17.0	15.1	12.3	10.8		
		MCDB	11.1	11.0	12.8	15.3	18.8	21.5	23.6	23.2	20.0	17.0	14.0	12.3		
	10%	WB	8.6	8.3	9.5	10.8	13.8	16.2	17.8	17.9	16.0	14.2	11.2	9.7		
		MCDB	10.1	10.0	11.8	14.0	17.2	20.0	22.2	22.0	18.8	16.1	12.9	11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	6.5	7.9	8.2	8.3	8.6	8.4	7.7	6.5	5.4	4.8		
		MCDBR	5.5	5.5	8.0	10.1	11.0	11.1	11.3	11.2	9.3	7.1	5.4	4.8		
	5% WB	MCWBR	5.1	4.5	5.4	6.1	6.5	6.2	5.5	5.4	5.2	5.2	4.6	4.5		
		MCWBR	5.5	5.3	7.4	9.6	10.1	10.3	10.2	10.1	8.3	6.6	5.3	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.349	0.386	0.407	0.413	0.414	0.411	0.402	0.381	0.365	0.337	0.323		
	taud		2.374	2.377	2.266	2.236	2.263	2.284	2.302	2.349	2.398	2.424	2.416	2.364		
	Ebn at Noon		671	768	800	829	842	842	837	826	801	739	660	614		
	Edh at Noon		63	81	108	125	128	126	122	111	95	77	60	54		
(44) All-Sky Solar Radiation	RadAvg		0.78	1.45	2.57	4.12	4.90	5.43	5.16	4.30	3.23	1.85	0.94	0.60		
	RadStd		0.07	0.22	0.32	0.39	0.35	0.42	0.46	0.33	0.26	0.18	0.11	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (80 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page